Lab – 07

Task-1:

```
PS D:\Data Structure\Lab Task7> cd "d:\Data Structure
Enter number of elements (at least 10): 10
Enter 10 elements:
1
2
5
4
7
5
9
7
8
6

 QUICK SORT

Quick Sort partition:
Selected pivot: 6
Left subarray: [1, 2, 5, 4, 5]
Pivot: 6
Right subarray: [9, 7, 8, 7]

Quick Sort partition:
Selected pivot: 5
Left subarray: [1, 2, 5, 4]
Pivot: 5
Right subarray: Empty

Quick Sort partition:
Selected pivot: 4
Left subarray: [1, 2]
Pivot: 4
Right subarray: [5]

Quick Sort partition:
Selected pivot: 2
Left subarray: [1]
Pivot: 2
Right subarray: Empty
```

```
Quick Sort partition:
Selected pivot: 7
Left subarray: [7]
Pivot: 7
Right subarray: [8, 9]

Quick Sort partition:
Selected pivot: 9
Left subarray: [8]
Pivot: 9
Right subarray: Empty

Final sorted array (Quick Sort): [1, 2, 4, 5, 5, 6, 7, 7, 8, 9]
Total subarrays generated: 9

 MERGE SORT

Dividing subarray: [1, 2, 5, 4, 7, 5, 9, 7, 8, 6]
Left subarray: [1, 2, 5, 4, 7]
Right subarray: [5, 9, 7, 8, 6]

Dividing subarray: [1, 2, 5, 4, 7]
Left subarray: [1, 2, 5]
Right subarray: [4, 7]

Dividing subarray: [1, 2, 5]
Left subarray: [1, 2]
Right subarray: [5]

Dividing subarray: [1, 2]
Left subarray: [1]
Right subarray: [2]
Merged subarray: [1, 2]
Merged subarray: [1, 2, 5]

Dividing subarray: [4, 7]
Left subarray: [4]
Right subarray: [7]
Merged subarray: [4, 7]
```

```
Merged subarray: [1, 2, 4, 5, 7]

Dividing subarray: [5, 9, 7, 8, 6]
Left subarray: [5, 9, 7]
Right subarray: [8, 6]

Dividing subarray: [5, 9, 7]
Left subarray: [5, 9]
Right subarray: [7]

Dividing subarray: [5, 9]
Left subarray: [5]
Right subarray: [9]
Merged subarray: [5, 9]
Merged subarray: [5, 7, 9]

Dividing subarray: [8, 6]
Left subarray: [8]
Right subarray: [6]
Merged subarray: [6, 8]
Merged subarray: [5, 6, 7, 8, 9]
Merged subarray: [1, 2, 4, 5, 5, 6, 7, 7, 8, 9]

Final sorted array (Merge Sort): [1, 2, 4, 5, 5, 6, 7, 7, 8, 9]
Total subarrays generated during division: 18
PS D:\Data Structure\Lab Task7>
```

Task-2:

```
PS D:\Data Structure\Lab Task7> cd "d:\Data Structure\Lab Task7\" ; if ($?) { g++ Q2.cpp -o Q2 } ;
Enter number of product IDs (at least 20): 20
Enter product IDs:
45 12 78 23 9 56 34 89 1 67 29 15 93 41 7 62 18 50 3 81

Original array: [45, 12, 78, 23, 9, 56, 34, 89, 1, 67, 29, 15, 93, 41, 7, 62, 18, 50, 3, 81]

Level 1
Technique: Quick Sort Partitioning
Selected pivot: 81
Left subarray: [45, 12, 78, 23, 9, 56, 34, 1, 67, 29, 15, 41, 7, 62, 18, 50, 3]
Pivot in final position: 81
Right subarray: [93, 89]
Subarrays generated at this split: 2

Level 2
Technique: Merge Sort Division
Left subarray: [45, 12, 78, 23, 9, 56, 34, 1, 67]
Right subarray: [29, 15, 41, 7, 62, 18, 50, 3]
Subarrays generated at this split: 2

Level 3
Technique: Quick Sort Partitioning
Selected pivot: 67
Left subarray: [45, 12, 23, 9, 56, 34, 1]
Pivot in final position: 67
Right subarray: [78]
Subarrays generated at this split: 2

Level 4
Technique: Merge Sort Division
Left subarray: [45, 12, 23, 9]
Right subarray: [56, 34, 1]
Subarrays generated at this split: 2

Level 5
Technique: Quick Sort Partitioning
Selected pivot: 9
Left subarray: Empty
```

```
Right subarray: [12, 23, 45]
Subarrays generated at this split: 1

Level 6
Technique: Merge Sort Division
Left subarray: [12, 23]
Right subarray: [45]
Subarrays generated at this split: 2

Level 7
Technique: Quick Sort Partitioning
Selected pivot: 23
Left subarray: [12]
Pivot in final position: 23
Right subarray: Empty
Subarrays generated at this split: 1
Merged subarray: [12, 23, 45]

Level 5
Technique: Quick Sort Partitioning
Selected pivot: 1
Left subarray: Empty
Pivot in final position: 1
Right subarray: [34, 56]
Subarrays generated at this split: 1

Level 6
Technique: Merge Sort Division
Left subarray: [34]
Right subarray: [56]
Subarrays generated at this split: 2
Merged subarray: [34, 56]
Merged subarray: [1, 9, 12, 23, 34, 45, 56]

Level 3
Technique: Quick Sort Partitioning
Selected pivot: 3
Left subarray: Empty
Pivot in final position: 3
```

```
Subarrays generated at this split: 1

Level 4
Technique: Merge Sort Division
Left subarray: [15, 41, 7, 62]
Right subarray: [18, 50, 29]
Subarrays generated at this split: 2

Level 5
Technique: Quick Sort Partitioning
Selected pivot: 62
Left subarray: [15, 41, 7]
Pivot in final position: 62
Right subarray: Empty
Subarrays generated at this split: 1

Level 6
Technique: Merge Sort Division
Left subarray: [15, 41]
Right subarray: [7]
Subarrays generated at this split: 2

Level 7
Technique: Quick Sort Partitioning
Selected pivot: 41
Left subarray: [15]
Pivot in final position: 41
Right subarray: Empty
Subarrays generated at this split: 1
Merged subarray: [7, 15, 41]

Level 5
Technique: Quick Sort Partitioning
Selected pivot: 29
Left subarray: [18]
Pivot in final position: 29
Right subarray: [50]
Subarrays generated at this split: 2
Merged subarray: [7, 15, 18, 29, 41, 50, 62]
```

```
Merged subarray: [1, 3, 7, 9, 12, 15, 18, 23, 29, 34, 41, 45, 50, 56, 62, 67, 78]

Level 2
Technique: Merge Sort Division
Left subarray: [93]
Right subarray: [89]
Subarrays generated at this split: 2
Merged subarray: [89, 93]

 LEVEL-WISE TOTALS
Level 1: 2 subarrays generated
Level 2: 4 subarrays generated
Level 3: 3 subarrays generated
Level 4: 4 subarrays generated
Level 5: 5 subarrays generated
Level 6: 6 subarrays generated
Level 7: 2 subarrays generated
Total subarrays generated: 26

Final sorted product IDs: [1, 3, 7, 9, 12, 15, 18, 23, 29, 34, 41, 45, 50, 56, 62, 67, 78, 81, 89, 93]
PS D:\Data Structure\Lab Task7>
```

Task-3:

```
PS D:\Data Structure\Lab Task7> cd "d:\Data Structure\Lab Task7\" ; if ($?) { g++ Q3.cpp -o Q3 } ; if ($?
Enter number of students (at least 20): 20
Enter the last four digits of each student ID (0000 to 9999):
4826 1357 2468 4826 7531
1234 5678 9012 3456 7890
1111 2222 3333 4444 5555
6666 7777 8888 9999 1234

Original IDs:
4826 1357 2468 4826 7531 1234 5678 9012 3456 7890 1111 2222 3333 4444 5555 6666 7777 8888 9999 1234

========== RADIX SORT ==========

After Units Pass:
7890 7531 1111 9012 2222 3333 1234 4444 1234 5555 4826 4826 3456 6666 1357 7777 2468 5678 8888 9999
IDs having an even digit at this position: 13

After Tens Pass:
1111 9012 2222 4826 4826 7531 3333 1234 1234 4444 5555 3456 1357 6666 2468 7777 5678 8888 7890 9999
IDs having an even digit at this position: 7

After Hundreds Pass:
9012 1111 2222 1234 1234 3333 1357 4444 3456 2468 7531 5555 6666 5678 7777 4826 4826 8888 7890 9999
IDs having an even digit at this position: 13

After Thousands Pass:
1111 1234 1234 1357 2222 2468 3333 3456 4444 4826 4826 5555 5678 6666 7531 7777 7890 8888 9012 9999
IDs having an even digit at this position: 7

========== DUPLICATE IDs ==========
ID: 1234 | Frequency: 2
ID: 4826 | Frequency: 2

========== DIGIT SUMS ==========
ID: 1111 | Digit Sum: 4
ID: 1234 | Digit Sum: 10
ID: 1234 | Digit Sum: 10
ID: 1357 | Digit Sum: 16
ID: 2222 | Digit Sum: 8
```

```
ID: 2468 | Digit Sum: 20
ID: 3333 | Digit Sum: 12
ID: 3456 | Digit Sum: 18
ID: 4444 | Digit Sum: 16
ID: 4826 | Digit Sum: 20
ID: 4826 | Digit Sum: 20
ID: 5555 | Digit Sum: 20
ID: 5678 | Digit Sum: 26
ID: 6666 | Digit Sum: 24
ID: 7531 | Digit Sum: 16
ID: 7777 | Digit Sum: 28
ID: 7890 | Digit Sum: 24
ID: 8888 | Digit Sum: 32
ID: 9012 | Digit Sum: 12
ID: 9999 | Digit Sum: 36

ID with highest digit sum: 9999
Highest digit sum: 36

========== LONGEST EVEN DIGIT-SUM SEQUENCE ==========
Longest sequence: 1111 1234 1234 1357 2222 2468 3333 3456 4444 4826 4826 5555 5678 6666 7531 7777 7890 8888 9012 9999
Sequence length: 20
PS D:\Data Structure\Lab Task7>
```

Task-4:

```
PS D:\Data Structure\Lab Task7> cd "d:\Data Structure\Lab Task7\" ; if
Enter the number of elements: 10
Enter decimal integers:
12 5 9 3 12 7 8 15 6 10

Binary Representations:
12 = 1100
5 = 0101
9 = 1001
3 = 0011
12 = 1100
7 = 0111
8 = 1000
15 = 1111
6 = 0110
10 = 1010

==================================
Pass 1 - Bit Position 0 (LSB)
==================================
Zeros: 5
Ones: 5
Array (Decimal): 12 12 8 6 10 5 9 3 7 15
Array (Binary): 1100 1100 1000 0110 1010 0101 1001 0011 0111 1111

==================================
Pass 2 - Bit Position 1
==================================
Zeros: 5
Ones: 5
Array (Decimal): 12 12 8 5 9 6 10 3 7 15
Array (Binary): 1100 1100 1000 0101 1001 0110 1010 0011 0111 1111

==================================
Pass 3 - Bit Position 2
==================================
Zeros: 4
Ones: 6
Array (Decimal): 8 9 10 3 12 12 5 6 7 15
```

```
Array (Binary): 1000 1001 1010 0011 1100 1100 0101 0110 0111 1111

====================================
Pass 4 - Bit Position 3 (MSB)
====================================
Zeros: 4
Ones: 6
Array (Decimal): 3 5 6 7 8 9 10 12 12 15
Array (Binary): 0011 0101 0110 0111 1000 1001 1010 1100 1100 1111

====================================
Final Sorted Array
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Decimal:
3 5 6 7 8 9 10 12 12 15

Binary:
0011 0101 0110 0111 1000 1001 1010 1100 1100 1111

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Additional Analysis
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Longest Binary Representation:
8 = 1000
Length = 4 bits

Largest Input Value:
15 = 1111
Number of set bits (1s): 4

Numbers with the same number of set bits:
15
Count: 1
PS D:\Data Structure\Lab Task7>
```